

Kinarm DDM parameter support matrix

June 23, 2026

Contents

Kinarm DDM parameter support matrix	2
Drift rate v	2
Practical reading of the evidence for v	3
Boundary separation a	3
Practical reading of the evidence for a	4
Nondecision time T_{er}	4
Practical reading of the evidence for T_{er}	6
Suggested starting support by modeling decision	6
Recommended starting parameter support summary	6
Most useful minimal citation set by parameter	7
References	8

Kinarm DDM parameter support matrix

This matrix is organized around the three main parameters of interest: drift rate v , boundary separation a , and nondecision time T_{er} . It separates four kinds of support that are often blurred together in methods writeups:

- **definition** of what the parameter means in a standard DDM
- **priors or bounds** that can be used in practice
- **task-specific constraints** from reaching, interception, saccades, and eye-hand work
- **cautions** about over-interpreting the fitted value

Drift rate v

Paper	What it supports for v	Type of support	Weight
(Ratcliff & McKoon, 2008)	Drift rate is the quality and direction of evidence accumulation in the standard DDM	Definition	Essential
(Ratcliff et al., 2016)	Absolute v values are not comparable across studies unless the scaling convention is matched	Interpretation caution	Essential
(Tran et al., 2020)	Provides the broad empirical envelope and prior source for v under $s = 1$ scaling	Prior and outer bound	Essential
(Wiecki et al., 2013)	Supports informative priors and hierarchical estimation of v	Estimation framework	High
(Vandekerckhove et al., 2011)	Supports shrinkage and stable subject-level estimation of v	Estimation framework	High
(Barany et al., 2020)	Suggests that in interception, fitted v may partly reflect urgency and overlap if the model is too simple	Task-specific caution	High
(Selen et al., 2012)	Shows evolving evidence can already shape the motor system before final commitment	Task-specific caution	Medium

Paper	What it supports for v	Type of support	Weight
(Dendauw et al., 2024)	In integrated decision-action models, evidence accumulation can influence later motor variables continuously rather than ending cleanly at one threshold crossing	Conceptual extension	Medium
(Purcell et al., 2010; Purcell et al., 2012)	Show neural accumulator accounts in which effective evidence readout is gated or architecture-dependent	Neural interpretation caution	Medium

Practical reading of the evidence for v

The literature supports using drift rate as a latent measure of evidence quality, but it does **not** support treating a fitted value of v as a task-independent biological speed limit (Ratcliff et al., 2016; Ratcliff & McKoon, 2008). For the Kinarm task, v should be bounded using the global review in (Tran et al., 2020) and interpreted conditionally on scaling and task design.

Boundary separation a

Paper	What it supports for a	Type of support	Weight
(Ratcliff & McKoon, 2008)	Boundary separation is the commitment criterion controlling the speed-accuracy tradeoff	Definition	Essential
(Ratcliff et al., 2016)	Absolute values of a depend on scaling and model specification	Interpretation caution	Essential
(Tran et al., 2020)	Provides the broad empirical envelope and prior source for a under $s = 1$ scaling	Prior and outer bound	Essential
(Wiecki et al., 2013)	Supports informative priors for hierarchical estimation of a	Estimation framework	High
(Vandekerckhove et al., 2011)	Supports stable hierarchical estimation of a across participants and conditions	Estimation framework	High

Paper	What it supports for a	Type of support	Weight
(Heitz & Schall, 2012)	Supports the view that threshold-like terms reflect policy and speed–accuracy setting rather than physiology	Conceptual support	High
(Cisek et al., 2009)	Suggests that in changing or urgent conditions, fixed boundaries may be a poor literal description	Task-specific caution	High
(Barany et al., 2020)	Moving-target interception likely increases urgency and may reduce the literal fit of a fixed-bound interpretation	Task-specific caution	High
(Dendauw et al., 2024)	Replaces a single boundary with a more layered gating and execution architecture in overlapping decision–movement settings	Conceptual extension	Medium

Practical reading of the evidence for a

The literature strongly supports boundary separation as a policy-like parameter for caution and commitment, but not as a physiological upper or lower bound in the same sense as motor delay (Heitz & Schall, 2012; Ratcliff & McKoon, 2008). For Kinarm data, a can be bounded using (Tran et al., 2020), but its interpretation should remain behavioral and strategic rather than anatomical.

Nondecision time T_{er}

Paper	What it supports for T_{er}	Type of support	Weight
(Ratcliff & McKoon, 2008)	Nondecision time is the portion of RT outside the decision process in the standard DDM	Definition	Essential
(Tran et al., 2020)	Provides a broad review-wide prior source for T_{er} , but one that is too wide for direct use in this task	Prior source	Essential
(Bompas et al., 2024)	Fitted T_{er} often absorbs model mismatch and should not be treated as a pure physiological delay	Main interpretation caution	Essential

Paper	What it supports for T_{er}	Type of support	Weight
(Weindel et al., 2020)	EMG shows motor time changes with speed–accuracy, stimulus strength, and correctness	Physiological caution	Essential
(Haith et al., 2016)	Reach preparation can occur around 130 ms while overt initiation is later by about 78 ms	Hand-specific lower-bound evidence	Essential
(Bompas et al., 2017)	Manual responses are much later than saccades mainly because of nondecisional delays outside the core selection process	Eye versus hand timing anchor	High
(Gopal et al., 2015)	Coordinated eye-hand behavior is consistent with a shared decision stage plus later hand-specific delay of about 160 ms	Eye-hand task-specific constraint	Essential
(Jana et al., 2017)	Eye-hand architecture can switch between common and separate accumulation depending on task context	Eye-hand interpretation caution	High
(Barany et al., 2020)	Interception encourages early movement launch and online correction, which can make fitted T_{er} absorb overlap between deciding and acting	Interception-specific caution	High
(Selen et al., 2012)	Motor preparation reflects evolving evidence before final commitment	Task-specific caution	Medium
(Dendauw et al., 2024)	Decision, preparation, and execution can overlap continuously, making a single T_{er} term only a coarse summary	Conceptual extension	High
(Verdonck & Tuerlinckx, 2016)	Useful for separating or rethinking nondecision-time treatment in RT models	Methods extension	Medium

Paper	What it supports for T_{er}	Type of support	Weight
(Ludwig et al., 2007)	Provides a sharper physiological account of saccadic dead time	Saccadic timing anchor	Medium

Practical reading of the evidence for T_{er}

This is the parameter where the evidence is both richest and most treacherous. The literature supports using T_{er} , constraining it with external timing evidence, and interpreting it cautiously. It does **not** support leaving T_{er} completely unconstrained and then treating the fitted value as a direct readout of pure encoding plus motor delay (Bompas et al., 2024; Weindel et al., 2020).

Suggested starting support by modeling decision

Modeling decision	Best supporting papers
Use a hierarchical DDM at all	(Ratcliff & McKoon, 2008; Vandekerckhove et al., 2011; Wiecki et al., 2013)
Use review-based priors for v and a	(Ratcliff et al., 2016; Tran et al., 2020)
State the scaling convention explicitly	(Ratcliff et al., 2016; Tran et al., 2020)
Constrain hand T_{er} more tightly than the global review prior	(Bompas et al., 2017; Bompas et al., 2024; Gopal et al., 2015; Haith et al., 2016)
Consider separate eye and hand output terms	(Bompas et al., 2017; Gopal et al., 2015; Jana et al., 2017)
Interpret moving-target fits more cautiously than stationary-target fits	(Barany et al., 2020; Dendauw et al., 2024; Selen et al., 2012)
Avoid overclaiming about across-trial variability parameters	(Böhm et al., 2018; Ratcliff & Tuerlinckx, 2002)

Recommended starting parameter support summary

Parameter	What can be claimed strongly	What should be claimed cautiously
Drift rate v	It is a standard latent evidence-quality parameter and can be given review-based outer bounds (Ratcliff & McKoon, 2008; Tran et al., 2020)	Its absolute magnitude is not a universal human limit and may shift with scaling and task architecture (Barany et al., 2020; Ratcliff et al., 2016)
Boundary separation a	It is a standard caution or commitment parameter and can be bounded with review-based priors (Ratcliff & McKoon, 2008; Tran et al., 2020)	It should not be described as a physiological response bound in the same sense as motor delay (Cisek et al., 2009; Heitz & Schall, 2012)

Parameter	What can be claimed strongly	What should be claimed cautiously
Nondecision time T_{er}	It should be constrained using reaching, saccadic, and eye-hand evidence (Bompas et al., 2017; Bompas et al., 2024; Gopal et al., 2015; Haith et al., 2016)	It should not be treated as a pure sensorimotor constant, especially in interception tasks (Barany et al., 2020; Dendauw et al., 2024; Weindel et al., 2020)

Most useful minimal citation set by parameter

- **Drift rate v :** (Ratcliff et al., 2016; Ratcliff & McKoon, 2008; Tran et al., 2020)
- **Boundary separation a :** (Heitz & Schall, 2012; Ratcliff & McKoon, 2008; Tran et al., 2020)
- **Nondecision time T_{er} :** (Barany et al., 2020; Bompas et al., 2024; Gopal et al., 2015; Haith et al., 2016; Weindel et al., 2020)

That minimal set is enough to justify the main claims in a methods section while keeping the citation load manageable.

References

- Barany, D. A., Gómez-Granados, A., Schroyer, M., Cutts, S. A., & Singh, T. (2020). Perceptual decisions about object shape bias visuomotor coordination during rapid interception movements. *Journal of Neurophysiology*. <https://doi.org/10.1152/jn.00098.2020>
- Böhm, U., Annis, J., Frank, M., Hawkins, G. E., Heathcote, A., Kellen, D., Kryptos, A.-M., Lerche, V., Logan, G., Palmeri, T., Ravenzwaaij, D. van, Servant, M., Singmann, H., Starns, J., Voss, A., Wiecki, T. V., Matzke, D., & Wagenmakers, E. (2018). Estimating across-trial variability parameters of the Diffusion Decision Model: Expert advice and recommendations. In *Journal of Mathematical Psychology*. <https://doi.org/10.1016/J.JMP.2018.09.004>
- Bompas, A., Hedge, C., & Sumner, P. (2017). Speeded saccadic and manual visuo-motor decisions: Distinct processes but same principles. *Cognitive Psychology*, 94, 26–52. <https://doi.org/10.1016/j.cogpsych.2017.02.002>
- Bompas, A., Sumner, P., & Hedge, C. (2024). Non-decision time: The Higgs Boson of decision. *Psychological Review*. <https://doi.org/10.1037/rev0000487>
- Cisek, P., Puşkaş, G., & El-Murr, S. (2009). Decisions in Changing Conditions: The Urgency-Gating Model. *The Journal of Neuroscience*, 29, 11560–11571. <https://doi.org/10.1523/JNEUROSCI.1844-09.2009>
- Dendauw, E., Evans, N. J., Logan, G. D., Haffen, E., Bennabi, D., Gajdos, T., & Servant, M. (2024). The Gated Cascade Diffusion Model: An Integrated Theory of Decision Making, Motor Preparation, and Motor Execution. *Psychological Review*, 131, 825–857. <https://doi.org/10.1037/rev0000464>
- Gopal, A., Viswanathan, P., & Murthy, A. (2015). A common stochastic accumulator with effector-dependent noise can explain eye-hand coordination. *Journal of Neurophysiology*, 113 7, 2033–2048. <https://doi.org/10.1152/jn.00802.2014>
- Haith, A., Pakpoor, J., & Krakauer, J. (2016). Independence of Movement Preparation and Movement Initiation. *The Journal of Neuroscience*, 36, 3007–3015. <https://doi.org/10.1523/JNEUROSCI.3245-15.2016>
- Heitz, R. P., & Schall, J. (2012). Neural Mechanisms of Speed-Accuracy Tradeoff. *Neuron*, 76, 616–628. <https://doi.org/10.1016/j.neuron.2012.08.030>
- Jana, S., Gopal, A., & Murthy, A. (2017). Evidence of common and separate eye and hand accumulators underlying flexible eye-hand coordination. *Journal of Neurophysiology*, 117 1, 348–364. <https://doi.org/10.1152/jn.00688.2016>
- Ludwig, C., Mildinhal, J., & Gilchrist, I. (2007). A population coding account for systematic variation in saccadic dead time. *Journal of Neurophysiology*, 97 1, 795–805. <https://doi.org/10.1152/JN.00652.2006>
- Purcell, B. A., Heitz, R. P., Cohen, J. Y., Schall, J., Logan, G. D., & Palmeri, T. (2010). Neurally Constrained Modeling of Perceptual Decision Making. *Psychological Review*, 117, 1113–1143. <https://doi.org/10.1037/a0020311>
- Purcell, B. A., Schall, J., Logan, G., & Palmeri, T. (2012). From Salience to Saccades: Multiple-Alternative Gated Stochastic Accumulator Model of Visual Search. *The Journal of Neuroscience*, 32, 3433–3446. <https://doi.org/10.1523/JNEUROSCI.4622-11.2012>
- Ratcliff, R., & McKoon, G. (2008). The Diffusion Decision Model: Theory and Data for Two-Choice Decision Tasks. *Neural Computation*, 20, 873–922. <https://doi.org/10.1162/neco.2008.12-06-420>

- Ratcliff, R., Smith, P. L., Brown, S. D., & McKoon, G. (2016). Diffusion Decision Model: Current Issues and History. *Trends in Cognitive Sciences*, 20, 260–281. <https://doi.org/10.1016/j.tics.2016.01.007>
- Ratcliff, R., & Tuerlinckx, F. (2002). Estimating parameters of the diffusion model: Approaches to dealing with contaminant reaction times and parameter variability. *Psychonomic Bulletin & Review*, 9, 438–481. <https://doi.org/10.3758/BF03196302>
- Selen, L., Shadlen, M., & Wolpert, D. (2012). Deliberation in the motor system: reflex gains track evolving evidence leading to a decision. *The Journal of Neuroscience : The Official Journal of the Society for Neuroscience*, 32, 2276–2286. <https://doi.org/10.1523/JNEUROSCI.5273-11.2012>
- Tran, N., Maanen, L. van, Heathcote, A., & Matzke, D. (2020). Systematic Parameter Reviews in Cognitive Modeling: Towards a Robust and Cumulative Characterization of Psychological Processes in the Diffusion Decision Model. *Frontiers in Psychology*, 11. <https://doi.org/10.3389/fpsyg.2020.608287>
- Vandekerckhove, J., Tuerlinckx, F., & Lee, M. (2011). Hierarchical diffusion models for two-choice response times. *Psychological Methods*, 16 1, 44–62. <https://doi.org/10.1037/a0021765>
- Verdonck, S., & Tuerlinckx, F. (2016). Factoring out nondecision time in choice reaction time data: Theory and implications. *Psychological Review*, 123 2, 208–218. <https://doi.org/10.1037/rev0000019>
- Weindel, G., Anders, R., Alario, F., & Burle, B. (2020). Assessing model-based inferences in decision making with single-trial response time decomposition. *Journal of Experimental Psychology. General*. <https://doi.org/10.31234/osf.io/kc26f>
- Wiecki, T. V., Sofer, I., & Frank, M. (2013). HDDM: Hierarchical Bayesian estimation of the Drift-Diffusion Model in Python. *Frontiers in Neuroinformatics*, 7. <https://doi.org/10.3389/fninf.2013.00014>